

MEDICAL INSURANCE COST

Statistical Analysis & Interactive Web Dashboard

Lab-4: Applied Statistical Modeling & Interactive Web Dashboard

Course	Statistical Modeling with Python
Course Code	DS602
Program	M.Sc. Data Science
Semester	1
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Dashboard	https://mysiteds.streamlit.app/

This document presents the statistical analysis, hypothesis testing, regression modeling, diagnostics, and interactive dashboard work completed for the Medical Insurance Costs dataset.

The supplied reference PDF was used only for structure and presentation; the statistical results here are specific to this submission.

1. Project Overview

This project applies statistical methods to the Medical Insurance Costs dataset and presents the results through an interactive Streamlit dashboard. The workflow covers data preparation, descriptive statistics, exploratory visualization, formal hypothesis testing, multiple linear regression, regression diagnostics, and live prediction.

Dashboard: <https://mysiteds.streamlit.app/>

2. Project Objectives

- Explore the structure and distribution of medical insurance cost data.
- Calculate mean, median, standard deviation, IQR, skewness, and kurtosis.
- Use histograms/KDE plots, scatter plots, and correlation analysis.
- Compare charges between smokers and non-smokers using assumption checks and a suitable test.
- Use One-Way ANOVA to compare average charges across regions.
- Fit a multiple linear regression model using statsmodels OLS.
- Interpret coefficients, p-values, confidence intervals, R^2 , and adjusted R^2 .
- Check residual behavior using diagnostic plots, Jarque-Bera, and VIF.
- Provide interactive exploration, testing, prediction, and diagnostics in Streamlit.

3. Dataset

The dataset originally contains 1,338 observations and 7 variables. One duplicate observation was removed, leaving 1,337 observations for the main analysis.

Variable	Type	Description
age	Numerical	Age of primary beneficiary
sex	Categorical	Sex of primary beneficiary
bmi	Numerical	Body Mass Index
children	Numerical	Number of children/dependents
smoker	Categorical	Smoking status
region	Categorical	Geographic region
charges	Numerical	Medical insurance charges

4. Technologies and Libraries

Tool	Purpose
Python	Core implementation
Pandas	Data handling, cleaning, filtering, summaries
NumPy	Numerical operations
SciPy	Statistical testing
Statsmodels	OLS regression and diagnostics
Matplotlib / Seaborn	EDA and diagnostic plots
Streamlit	Interactive dashboard
Git / GitHub	Version control and submission

5. Project Structure

```
202618063_DS602/  
■■■ data/insurance.csv  
■■■ models/  
■■■ analysis.ipynb  
■■■ app.py  
■■■ requirements.txt  
■■■ README.md  
■■■ Lab4_DS602_MSc.pdf
```

6. Exploratory Data Analysis

6.1 Data Preparation

- Original size: 1,338 rows × 7 columns.
- Missing values: none.
- One duplicate was found and removed.
- Final analysis size: 1,337 rows × 7 columns.
- Numerical variables: age, bmi, children, charges.
- Categorical variables: sex, smoker, region.

6.2 Descriptive Statistics

Variable	Mean	Median	SD	IQR	Skewness	Kurtosis
age	39.22	39.00	14.04	24.00	0.05	-1.24
bmi	30.66	30.40	6.10	8.41	0.28	-0.05
children	1.10	1.00	1.21	2.00	0.94	0.20
charges	13279.12	9386.16	12110.36	11911.37	1.52	1.60

- Age is approximately balanced because mean and median are close and skewness is 0.05.
- BMI has a small positive skew of 0.28.
- Children has moderate positive skewness of 0.94.
- Charges show strong positive skewness of 1.52; high-cost observations pull the distribution to the right.

6.3 Visual Exploration

- Histograms with KDE were used to inspect numerical distributions.
- A BMI versus charges scatter plot was used to study the relationship between BMI and insurance cost, with smoking status shown by group.
- A correlation heatmap was used to examine linear relationships among numerical variables.

7. Correlation Analysis

	age	bmi	children	charges
age	1.00	0.11	0.04	0.30
bmi	0.11	1.00	0.01	0.20
children	0.04	0.01	1.00	0.07
charges	0.30	0.20	0.07	1.00

The strongest simple numerical relationship with charges is age ($r = 0.30$), followed by BMI ($r = 0.20$). Children has a weak correlation with charges ($r = 0.07$).

8. Hypothesis Testing

8.1 Two-Group Comparison: Smoker vs Non-Smoker Charges

H0: There is no significant difference in charges between smokers and non-smokers.

H1: There is a significant difference.

α : 0.05.

Test / Group	Statistic	p-value	Decision
Smokers	n = 274	—	Mean = 32050.23
Non-smokers	n = 1063	—	Mean = 8440.66
Shapiro-Wilk: smoker	0.9396	3.62×10^{-13}	Reject normality
Shapiro-Wilk: non-smoker	0.8729	1.50×10^{-13}	Reject normality
Levene	332.4714	1.67×10^{-13}	Reject equal variance
Mann-Whitney U	283859.0	5.75×10^{-13}	Reject H0

Conclusion: Both groups fail the normality check, so the Mann-Whitney U test is used. Its p-value is far below 0.05, so H0 is rejected. Insurance charges differ significantly between smokers and non-smokers, with smokers having much higher mean charges.

8.2 One-Way ANOVA: Charges Across Regions

H0: Regional mean charges are equal.

H1: At least one regional mean differs.

α : 0.05.

Region	Count	Mean Charges
northeast	324	13406.38
northwest	324	12450.84
southeast	364	14735.41
southwest	325	12346.94
ANOVA	F = 2.9261	p = 0.03276

Conclusion: $p = 0.03276 < 0.05$, so H0 is rejected. Average insurance charges are not equal across all regions. A post-hoc test would be needed to identify the exact regional pairs that differ.

9. Multiple Linear Regression

A multiple linear regression model was fitted with charges as the response. Numerical predictors were age, BMI, and children, while categorical predictors were dummy encoded.

Model form: $\text{charges} = \beta_0 + \beta_1(\text{age}) + \beta_2(\text{bmi}) + \beta_3(\text{children}) + \text{categorical effects} + \epsilon$

Predictor	Coefficient	p-value	95% CI	Result
Intercept	-11940 approx.	<0.001	(-13900,-9997.9)	Significant
age	256.7646	<0.001	(233.396,280.133)	Significant
bmi	339.2504	<0.001	(283.122,395.379)	Significant
children	474.8205	0.001	(204.301,745.340)	Significant
sex_male	-129.4815	0.698	(-783.128,524.165)	Not significant
smoker_yes	23850 approx.	<0.001	approx. (23000,24700)	Significant
region_northwest	-349.2265	0.464	(-1284.637,586.183)	Not significant
region_southeast	-1035.2656	0.031	(-1974.684,-95.847)	Significant
region_southwest	-960.0814	0.045	(-1898.007,-22.156)	Significant

Model fit

- $R^2 = 0.751$, so approximately 75.1% of the variation in charges is explained by the fitted predictors.
- Adjusted $R^2 = 0.749$, remaining close to R^2 after accounting for model complexity.
- F-statistic = 500.0, indicating a statistically significant overall model.

Coefficient interpretation

- Age: each additional year is associated with about 256.76 higher charges, holding other predictors constant.
- BMI: each one-unit increase is associated with about 339.25 higher charges.
- Children: each additional child/dependent is associated with about 474.82 higher charges.
- Smoker status has the strongest positive effect in the fitted model.
- Sex is not statistically significant, while southeast and southwest have small significant negative effects relative to the reference region.

10. Regression Diagnostics

10.1 Residual Normality

The OLS output reports an Omnibus statistic of 299.816 and a Jarque-Bera statistic of 716.5524. The Jarque-Bera p-value is approximately 2.53×10^{-15} .

Because the p-value is far below 0.05, the null hypothesis of normally distributed residuals is rejected. The residuals therefore show a clear departure from normality.

10.2 Variance Inflation Factor

VIF is used to assess multicollinearity among continuous predictors. The dashboard includes VIF diagnostics for variables such as age, BMI, and children.

10.3 Additional Diagnostic Information

- Residual-vs-fitted values help inspect linearity and changing variance.
- The Q-Q plot helps inspect approximate residual normality.
- Durbin-Watson = 2.089 in the OLS output, which does not indicate strong first-order residual autocorrelation.

11. Interactive Streamlit Dashboard

Dashboard: <https://mysiteds.streamlit.app/>

Dashboard Area	Main Features	Purpose
Tab 1 – Data Exploration	Filters, summaries, distributions, scatter plots, correlation	Interactive EDA
Tab 2 – Hypothesis Testing Lab	Group selection, statistics, p-values, conclusions	Interactive inference
Tab 3 – Live Prediction & Diagnostics	Inputs, prediction, intervals, diagnostics	Prediction and validation

12. Main Statistical Findings

- The final dataset contains 1,337 observations after removing one duplicate.
- Charges are strongly right-skewed, with skewness of 1.52.
- Age has the strongest simple numerical correlation with charges among the numerical predictors ($r = 0.30$).
- Smokers have substantially higher mean charges than non-smokers (32050.23 vs 8440.66).
- The Mann-Whitney U test confirms a statistically significant difference between smoker and non-smoker charge distributions.
- One-Way ANOVA indicates a statistically significant difference in average charges across regions.
- The OLS model explains about 75.1% of the observed variation in charges.
- Age, BMI, children, and smoker status are statistically significant predictors in the fitted model, while sex is not.
- Residual normality is not supported by the Jarque-Bera test.

13. Reproducibility

- Open the project folder in VS Code.
- Verify that data/insurance.csv is present.
- Install the packages listed in requirements.txt.
- Launch the dashboard using: `streamlit run app.py`.
- Use the three dashboard areas to explore, test, model, and predict.

14. Version Control and Submission

The project is maintained in the GitHub repository for the DS602 submission. The repository contains the analysis notebook, Streamlit application, dataset folder, model-related files, dependency file, README, and academic documentation.

15. Limitations

- The data are observational, so association does not establish causation.
- Medical charges may depend on factors not represented in the dataset.
- Categorical reference coding affects regression coefficient interpretation.
- OLS inference depends on assumptions that should be assessed using diagnostics.
- Predictions and intervals are statistical estimates and contain uncertainty.

16. Conclusion

This project demonstrates an end-to-end statistical workflow using the Medical Insurance Costs dataset. It integrates data preparation, descriptive statistics, visualization, formal hypothesis testing, multiple linear regression, diagnostic checking, and interactive prediction in a Streamlit application. The results show a strong association between smoking status and insurance charges, statistically significant regional differences, and a regression model that explains a substantial portion of charge variability.

Reference note: the supplied Documentation_DS602.pdf was used only as a reference for document organization and presentation, not as a source for the numerical results in this submission.